

Total No. of Questions :5]

SEAT No. :

PE4199

[Total No. of Pages : 2

[6554]-2003

F.Y.M.C.A. (Management)

STQ-553-MJ: SOFTWARE TESTING & QUALITY ASSURANCE

(2024 Pattern) (Semester-II)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*
- 3) *Draw diagrams wherever necessary.*

Q1) a) Describe software Quality Assurance and its challenges. **[5]**

b) Illustrate ISO 9000 standards in SQA **[5]**

OR

c) Distinguish between software quality assurance and software quality control. **[5]**

d) Define Reliability and explain the measures Availability, ROCOF, POFOD **[5]**

Q2) a) Explain the Testing Principles. **[5]**

b) Illustrate V model phases. **[5]**

OR

c) Discuss any five non-functional testing types. **[5]**

d) Differentiate between Re-testing and Regression Testing. **[5]**

Q3) a) You are asked with testing an Age Verification system that accepts the age of a person. The system allows users who are between 18 and 60 years old to access certain features.

Using the Boundry value Analysis technique. **[5]**

i) Determine the number of test cases required to test the above functionality.

ii) Specify the test cases, including representative values and their expected outcomes.

b) Describe static analysis tools and explain its importance. **[5]**

OR

P.T.O.

- c) For the given code, calculate the cyclomatic complexity. [5]

Read A

If A>0 Then

If A=21 Then

Print "Key"

End If

End If

- d) Differentiate between black box and white box testing. [5]

Q4) An online learning platform, Knowledge provides interactive courses, video lectures and quizzes. It supports user roles such as students, instructors and administrators. The platform includes features like personalized learning recommendations, discussion forum, progress tracking and integration with third party tools like payment gateways.

- a) Write a test plan for the following sections of the IEEE 829 test plan template for the above application. [10]

- i) Scope of testing
- ii) Objectives
- iii) Risks
- iv) Strategy
- v) Approach

OR

- b) Write suitable test cases for the above application. [10]

Q5) a) How can a testing tool be successfully introduced and integrated into an organizations workflow? [5]

- b) Discuss the selenium components-WebDriver and TestNG. [5]

OR

- c) Explain the benefits of JIRA in software Test management. [5]

- d) Narrate the stages of Defect Life cycle with a diagram. [5]

